

RNA-Seq Pipeline Master Runbook

Project: PRJNA229998 Airway RNA-Seq Pipeline **Environment:** Slurm HPC Cluster

1. Pipeline Architecture Overview

This pipeline is designed for high-reproducibility and robust error-handling on Slurm-managed High-Performance Computing (HPC) clusters. It features a two-tiered architecture: 1. **Modular Suite (Scripts 1-7, 9):** Step-by-step execution for development, debugging, and validation. 2. **V9 Monolithic Suite (Script 8):** The production-grade, automated loop for processing entire sample cohorts continuously.

2. Crucial HPC Safeguards (The "V9" Fixes)

During validation, three critical stability patches were implemented globally: *

Memory Allocation (`std::bad_alloc`): STAR indexing requires massive memory. All STAR batch scripts explicitly request `#SBATCH --mem=36G`. *

Read-Pair Synchronization: Fastq dumping uses `fasterq-dump --split-3` to ensure empty/dropped reads do not break Read 1 / Read 2 parity. *

Environment Isolation: Every script issues `module purge` prior to loading toolchains to prevent hidden cluster library conflicts. *

Dynamic Bioconductor Setup: Rsubread is automatically bootstrapped into a local `R_libs` scratch directory, bypassing administrative cluster blocks.

3. Directory Structure

```
/scratch/$(whoami)/
└─ PRJNA229998_airway_pipeline_ref/          # Shared STAR Index & GTF References
```

```
└─ PRJNA229998_airway_pipeline_stepbystep/ # Modular validation workspace
└─ PRJNA229998_airway_pipeline_automated/  # V9 Cohort production workspace
    └─ raw_reads/
    └─ trimmed_reads/
    └─ alignments/
    └─ counts/
    └─ R_libs/
```

4. Execution Procedures

A. Running the Automated Cohort (Production)

1. Submit the monolithic script: `sbatch 8_Automated_Pipeline_AllSamples.sh`
2. Monitor progress: `tail -f $(ls -t ~/slurm-*.out | head -n 1)`

B. Running Modular Validation (Debugging)

Execute in strict sequence, ensuring the previous job finishes before submitting the next: 1. `sbatch 1_Fetch_SRA.sh` 2. `sbatch 2_FastQC_Raw.sh` 3. `sbatch 3_Trim_fastp.sh` 4. `sbatch 4_FastQC_Clean.sh` 5. `sbatch 5_STAR_Index.sh` (*Run once per reference genome*) 6. `sbatch 6_STAR_Align.sh` 7. `sbatch 7_Quantification_Rsubread.sh` 8. `bash 9_QC_Report_Generator.sh` (*Generates final Markdown report*)

5. Monitoring & Troubleshooting

- **Check Queue:** `squeue -u $(whoami)`
 - **Cancel Job:** `scancel <JOB_ID>`
 - **Search for Errors:** `grep -iE "error|fatal|cannot|execution halted" ~/slurm-*.out`
 - **Verify STAR Alignment:** `tail -n 20 alignments/*_Log.final.out`
 - **Verify Count Assignments:** `cat counts/*_counts.txt.summary`
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6. Downstream Handoff

The ultimate output is the raw count matrix (`counts/airway_raw_counts.txt` or `*_counts.txt`). This tab-delimited file is formatted with Gene IDs as rows and Sample IDs as columns, ready for direct import into **DESeq2** in R for differential gene expression analysis.